AQA A-Level Further Mathematics (7367) — Topic 1.2

1.2: Matrices

1. Introduction to Matrices

Matrix: A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns. The entry in row i, column j is denoted a_{ij}.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Special Matrices

- **Identity matrix I:** Square matrix with 1s on the diagonal, 0s elsewhere. $AI = IA = A$.
- **Zero matrix O:** All entries are 0. $A + O = A$.
- **Transpose A^T:** Rows and columns swapped. $(A^T)_{ij} = A_{ji}$.
- **Symmetric matrix:** $A^T = A$.

2. Matrix Operations

Addition and Scalar Multiplication

Matrices must be the same size to add. Add corresponding entries. Scalar multiplication multiplies every entry:

$$(A+B)_{ij} = a_{ij} + b_{ij} \qquad (kA)_{ij} = k\,a_{ij}$$

Matrix Multiplication

For A of size $m \times p$ and B of size $p \times n$, the product AB is $m \times n$:

$$(AB)_{ij} = \sum_{k=1}^{p} a_{ik}\,b_{kj}$$

Row i of A dotted with column j of B.

Example: Compute AB where $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

$$AB = \begin{pmatrix} 1\cdot 5 + 2\cdot 7 & 1\cdot 6 + 2\cdot 8 \\ 3\cdot 5 + 4\cdot 7 & 3\cdot 6 + 4\cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

AQA Exam Tip: Matrix multiplication is NOT commutative: $AB \neq BA$ in general. Always check the order. AQA may test this explicitly.

Properties of Matrix Multiplication

- Associativity: $(AB)C = A(BC)$
- Distributivity: $A(B+C) = AB + AC$
- $(AB)^T = B^T A^T$
- $AI = IA = A$

3. Determinants

2×2 Determinant

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 Determinant (Cofactor Expansion)

Expanding along the first row:

$$\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Example: Find $\det\begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & -2 \end{pmatrix}$.

$$= 2[(-1)(-2) - (4)(2)] - 1[(0)(-2) - (4)(1)] + 3[(0)(2) - (-1)(1)]$$
$$= 2[2 - 8] - 1[0 - 4] + 3[0 + 1]$$
$$= 2(-6) - 1(-4) + 3(1) = -12 + 4 + 3 = -5$$

Properties of Determinants

- $\det(AB) = \det(A)\det(B)$
- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix
- Swapping two rows changes the sign of the determinant
- If two rows are identical, $\det = 0$

4. Inverse Matrices

2×2 Inverse

$$A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{where } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},\ \det A \neq 0$$

Example: Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$.

$\det A = 6 - 5 = 1$
$A^{-1} = \frac{1}{1}\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$

3×3 Inverse via Cofactors

The inverse of a 3×3 matrix A with $\det A \neq 0$ is:

$$A^{-1} = \frac{1}{\det A}(\operatorname{adj} A) \quad \text{where } \operatorname{adj} A = (C_{ij})^T$$

Here $C_{ij} = (-1)^{i+j} M_{ij}$ is the cofactor (signed minor) of element a_{ij}.

Example: Find A^{-1} for $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 0 \end{pmatrix}$.

$\det A = 1(0-3) - 2(0-3) + 1(0-1) = -3 + 6 - 1 = 2$

Cofactors: $C_{11} = +(0-3) = -3$, $C_{12} = -(0-3) = 3$, $C_{13} = +(0-1) = -1$,
$C_{21} = -(0-1) = 1$, $C_{22} = +(0-1) = -1$, $C_{23} = -(1-2) = 1$,
$C_{31} = +(6-1) = 5$, $C_{32} = -(3-0) = -3$, $C_{33} = +(1-0) = 1$

$\operatorname{adj} A = \begin{pmatrix} -3 & 1 & 5 \\ 3 & -1 & -3 \\ -1 & 1 & 1 \end{pmatrix}$, so $A^{-1} = \frac{1}{2}\begin{pmatrix} -3 & 1 & 5 \\ 3 & -1 & -3 \\ -1 & 1 & 1 \end{pmatrix}$

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AQA Exam Tip: The sign pattern for cofactors is $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$. Always verify your inverse by checking $AA^{-1} = I$ if time permits.

5. Solving Simultaneous Equations with Matrices

The system $A\mathbf{x} = \mathbf{b}$ has solution $\mathbf{x} = A^{-1}\mathbf{b}$ when $\det A \neq 0$.

Example: Solve $\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases}$.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \det A = -2 - 1 = -3$$
$$A^{-1} = -\frac{1}{3}\begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$$

$$\mathbf{x} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 5 \\ 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

So $x = 2$, $y = 1$.

Singular Matrices and System Analysis

When $\det A = 0$, the matrix is **singular** and has no inverse. The system $A\mathbf{x} = \mathbf{b}$ is either:

- **Inconsistent** — no solutions (parallel planes that don't all meet)
- **Consistent with infinitely many solutions** — the equations are dependent

AQA Exam Tip: For a 3×3 system, if $\det A = 0$, you must determine whether the system is inconsistent or has infinitely many solutions. Substitute a general value (e.g. let $z = t$) and check consistency of the remaining equations.

6. Linear Transformations in 2D

A 2×2 matrix A represents a linear transformation of the plane. The columns of A are the images of the standard basis vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Standard 2D Transformations

Transformation	Matrix	Notes
Rotation by θ (anticlockwise about origin)	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	det $= 1$
Reflection in x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	det $= -1$
Reflection in y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	det $= -1$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	det $= -1$
Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	det $= -1$
Enlargement, scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	det $= k^2$
Stretch, factor k in x-direction	$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$	det $= k$
Shear, factor k parallel to x-axis	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	det $= 1$

Area scale factor: Under the transformation represented by matrix A, areas scale by $|\det A|$. Orientation is preserved if $\det A > 0$ and reversed if $\det A < 0$.

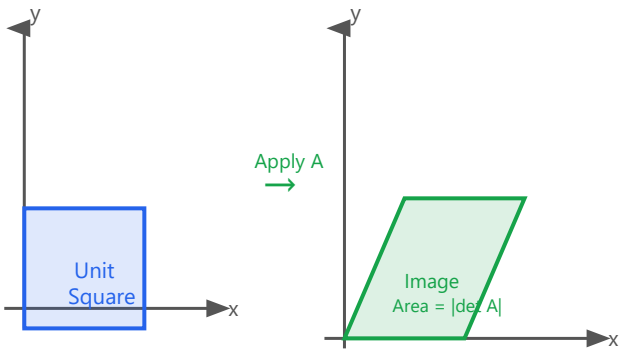
Composition of Transformations

Applying transformation B first, then transformation A: the combined matrix is AB (NOT BA). Read from right to left.

Example: The unit square has vertices at $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$. Under the transformation $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, find the image vertices and the area of the image.

Images: $\begin{pmatrix} 0 \\ 0 \end{pmatrix} \to \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \to \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \to \begin{pmatrix} 3 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix} \to \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\det A = 6 - 0 = 6$. Area of image $= |\det A| \times 1 = 6$.



7. Linear Transformations in 3D

A 3×3 matrix represents a linear transformation in 3D space. Key examples:

Transformation	Matrix
Rotation by θ about z-axis	$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Reflection in xy-plane ($z \to -z$)	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
Enlargement, scale factor k	$k\,I_3$

For 3D transformations, volumes scale by $|\det A|$.

8. Eigenvalues and Eigenvectors

Eigenvalue and eigenvector: A non-zero vector $\mathbf{v}$ is an *eigenvector* of matrix A with corresponding *eigenvalue* λ if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Geometrically: the transformation A maps $\mathbf{v}$ to a scalar multiple of itself (same or opposite direction).

Finding Eigenvalues: Characteristic Equation

Rearranging: $(A - \lambda I)\mathbf{v} = \mathbf{0}$. For non-zero $\mathbf{v}$, the matrix $(A - \lambda I)$ must be singular:

$$\det(A - \lambda I) = 0 \quad \text{(characteristic equation)}$$

Example: Find the eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

$$\det(A - \lambda I) = \det \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0$$

$(\lambda - 2)(\lambda - 5) = 0 \Rightarrow \lambda = 2$ or $\lambda = 5$.

Finding Eigenvectors

For each eigenvalue λ, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

Example (continued): Find eigenvectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

For $\lambda = 2$: $(A - 2I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$

$2x + y = 0 \Rightarrow y = -2x$. Eigenvector: $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ (or any scalar multiple).

For $\lambda = 5$: $(A - 5I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0}$

$-x + y = 0 \Rightarrow y = x$. Eigenvector: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

AQA Exam Tip: Eigenvectors are not unique — any non-zero scalar multiple is equally valid. AQA mark schemes accept any correct scalar multiple. State clearly which eigenvalue corresponds to which eigenvector.

Eigenvalues and Diagonalisation (Further Context)

If A has linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots$ with eigenvalues $\lambda_1, \lambda_2, \ldots$, then forming $P = [\mathbf{v}_1 \ \mathbf{v}_2 \ \ldots]$ gives $P^{-1}AP = D$ (diagonal), where $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$. This is used in solving certain differential equations and computing matrix powers.

9. Cayley-Hamilton Theorem

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation. That is, if $\det(\lambda I - A) = \lambda^2 + p\lambda + q$ (for a 2×2 matrix), then $A^2 + pA + qI = O$.

Example: For $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, the characteristic equation is $\lambda^2 - 7\lambda + 10 = 0$.

By Cayley-Hamilton: $A^2 - 7A + 10I = O$.

Verify: $A^2 = \begin{pmatrix} 18 & 7 \\ 14 & 11 \end{pmatrix}$, $7A = \begin{pmatrix} 28 & 7 \\ 14 & 21 \end{pmatrix}$

$A^2 - 7A + 10I = \begin{pmatrix} 18 - 28 + 10 & 7 - 7 + 0 \\ 14 - 14 + 0 & 11 - 21 + 10 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ ✓

The Cayley-Hamilton theorem can be used to find A^{-1} or to express higher powers of A in terms of lower powers.

Using Cayley-Hamilton to find A^{-1}: From $A^2 - 7A + 10I = O$, rearranging:
$10I = 7A - A^2 = A(7I - A)$
$A^{-1} = \frac{1}{10}(7I - A) = \frac{1}{10}\begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix}$

10. Summary of Key Results

Result	Formula / Statement
2×2 determinant	$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
2×2 inverse	$A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
$\det(AB)$	$\det(A)\det(B)$
Area scale factor	$\lvert\det A\rvert$
Characteristic equation	$\det(A - \lambda I) = 0$
Eigenvector equation	$A\mathbf{v} = \lambda\mathbf{v}$
Cayley-Hamilton	Matrix satisfies its own characteristic equation
Rotation matrix	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

11. Practice Checklist

- Can you multiply two 3×3 matrices without errors?
- Can you compute the determinant of a 3×3 matrix by cofactor expansion?
- Can you find the inverse of a 2×2 and 3×3 matrix?
- Can you use matrices to solve 3×3 simultaneous equations?
- Can you identify and write down the standard 2D transformation matrices?

- Can you find eigenvalues and eigenvectors of a 2×2 or 3×3 matrix?
- Can you apply the Cayley-Hamilton theorem to find A^{-1} or simplify matrix expressions?

Eigenvalues and Eigenvectors — Full 3×3 Worked Example

For a 3×3 matrix, the process is the same as 2×2: find $\det(A - \lambda I) = 0$, then solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue.

Example: Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Step 1 — Characteristic equation:

$$\det(A - \lambda I) = \det \begin{pmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{pmatrix}$$

Expanding along the third column: $(3-\lambda)\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (3-\lambda)[(2-\lambda)^2 - 1]$.

$$= (3-\lambda)(\lambda^2 - 4\lambda + 3) = (3-\lambda)(\lambda-1)(\lambda-3) = -((\lambda-3)^2(\lambda-1))$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$ (repeated).

Step 2 — Eigenvectors:

$\lambda = 1$: $A - I = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. Row reduce: gives $x + y = 0$, $z = 0$. Eigenvector: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$.

$\lambda = 3$: $A - 3I = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Row reduce: gives $x = y$, z free. Two linearly independent eigenvectors: $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

Cayley-Hamilton Theorem

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation. If $p(\lambda) = \det(\lambda I - A)$, then $p(A) = 0$ (the zero matrix).

Proof sketch (2×2): For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the characteristic polynomial is $p(\lambda) = \lambda^2 - (a+d)\lambda + (ad - bc)$. Compute $A^2 - (a+d)A + (ad-bc)I$ directly to verify it equals $\mathbf{0}$. [Full proof uses the adjugate matrix.]

Applications:

- Finding A^{-1}: if $A^2 - \mathrm{tr}(A) \cdot A + \det(A) \cdot I = 0$, then $A^{-1} = \frac{1}{\det A}(\mathrm{tr}(A) \cdot I - A)$.
- Reducing powers: express A^n as a linear combination of I and A (for 2×2) using the characteristic equation repeatedly.

Example: $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$. Char poly: $\lambda^2 - 5\lambda + 4 = 0$. By Cayley-Hamilton: $A^2 - 5A + 4I = 0$, so $A^2 = 5A - 4I$. Then $A^3 = A \cdot A^2 = 5A^2 - 4A = 5(5A - 4I) - 4A = 21A - 20I$, etc.

Diagonalisation

If an $n \times n$ matrix A has n linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, then A is **diagonalisable**:

$$P^{-1}AP = D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

where $P = [\mathbf{v}_1 | \mathbf{v}_2 | \cdots | \mathbf{v}_n]$ (columns are eigenvectors).

Computing A^n: From $A = PDP^{-1}$ it follows that $A^n = PD^nP^{-1}$, and $D^n = \mathrm{diag}(\lambda_1^n, \ldots, \lambda_n^n)$ is trivial to compute.

Example: $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Eigenvalues $\lambda = 1, 3$; eigenvectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$A^n = P\begin{pmatrix}1^n & 0\\ 0 & 3^n\end{pmatrix}P^{-1} = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 3^n\end{pmatrix}\begin{pmatrix}1 & -1\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 3^n-1\\ 0 & 3^n\end{pmatrix}$$

Geometric Interpretation of the Determinant

For a 2×2 matrix A:

- The **signed area** of the image of the unit square under A equals $\det(A)$.
- If $\det(A) > 0$, orientation is preserved; if $\det(A) < 0$, orientation is reversed (reflected).
- If $\det(A) = 0$, the transformation collapses the plane to a line or point (rank ≤ 1).

For a 3×3 matrix, $|\det(A)|$ gives the factor by which volumes are scaled. The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \det[\mathbf{a}\,\mathbf{b}\,\mathbf{c}]$ gives the signed volume of the parallelepiped.

AQA Exam Technique — Matrices:

- **Eigenvalues:** Always fully factorise the characteristic polynomial. For 3×3, try $\lambda = \mathrm{tr}(A)/3$ as a first guess, or look for a row that gives a factor.
- **Eigenvectors:** After finding λ, row-reduce $A - \lambda I$ carefully — set one variable as the parameter.
- **Cayley-Hamilton:** Show explicitly that $p(A) = \mathbf{0}$ by computing A^2, then combining.
- **Diagonalisation:** State P, D, and confirm the order matches (column k of P corresponds to diagonal entry k of D).
- **Transformation composition:** ST means apply T first, then S. Read right to left.
- **Mark scheme:** Correct A^{-1} $[M1]$, correct solution $[A1]$; wrong method but correct determinant $[M1]$ if using Cramer's rule.

Standard Transformation Matrices (2D)

Transformation	Matrix	Notes
Rotation by θ anticlockwise	$\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$	det = 1, preserves area

Reflection in x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	det = −1
Reflection in y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	det = −1
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	det = −1
Reflection in $y = -x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	det = −1
Reflection in $y = x \tan \alpha$	$\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$	det = −1
Stretch by a in x, b in y	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	det = ab
Enlargement scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$	det = k^2
Shear: x-direction, factor k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$	det = 1

Solving 3×3 Systems — Row Reduction

Given $A\mathbf{x} = \mathbf{b}$, form the augmented matrix $[A|\mathbf{b}]$ and row reduce to echelon form.

Example: Solve $x + 2y + z = 4$, $2x + y - z = 1$, $3x - y + 2z = 5$.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 3 & -1 & 2 & 5 \end{array}\right)$$

$R_2 \to R_2 - 2R_1$: $(0, -3, -3| -7)$. $R_3 \to R_3 - 3R_1$: $(0, -7, -1| -7)$. $R_3 \to 3R_3 - 7R_2$: $(0, 0, 20|0)$. So $z = 0$. From R_2: $-3y - 3(0) = -7 \Rightarrow y = 7/3$. From R_1: $x = 4 - 2(7/3) - 0 = 4 - 14/3 = 2/3$. Solution: $x = 2/3$, $y = 7/3$, $z = 0$.

The Three Inconsistency Cases for 3×3 Systems

After row reduction, the augmented matrix reveals the nature of the solution:

Reduced form	Interpretation
Full rank 3: $\left(\begin{array}{ccc\|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right)$	Unique solution
Rank 2, consistent: bottom row $(0, 0, 0 \| 0)$	Infinite solutions — line in $\mathbb{R}^3$
Rank 2, inconsistent: bottom row $(0, 0, 0 \| k)$ with $k \neq 0$	No solution (triangular prism)
Two rows proportional (rank 1 only): but inconsistent	No solution (parallel planes)

Properties of Eigenvalues

- **Trace:** $\mathrm{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ (sum of eigenvalues = sum of diagonal entries)

  $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$ (product of eigenvalues)
- **Powers:** If $A\mathbf{v} = \lambda\mathbf{v}$, then $A^k\mathbf{v} = \lambda^k\mathbf{v}$
- **Inverse:** If A is invertible with eigenvalue λ, then A^{-1} has eigenvalue $1/\lambda$
- **Transpose:** A and A^T have the same eigenvalues (but generally different eigenvectors)
- **Symmetric matrices:** Always have real eigenvalues and orthogonal eigenvectors

Using trace and determinant to find eigenvalues of a 2×2: For $A = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix}$: $\mathrm{tr}(A) = 9$, $\det(A) = 20 - 2 = 18$. Characteristic equation: $\lambda^2 - 9\lambda + 18 = 0 \Rightarrow (\lambda - 3)(\lambda - 6) = 0$. So $\lambda_1 = 3$, $\lambda_2 = 6$. Check: $3 + 6 = 9 = \mathrm{tr}(A)$ ✓, $3 \times 6 = 18 = \det(A)$ ✓.

Worked Cayley-Hamilton Application

Find A^{-1} using Cayley-Hamilton for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$. Characteristic polynomial: $\lambda^2 - 7\lambda + 10 = 0$. By Cayley-Hamilton: $A^2 - 7A + 10I = 0$. Rearranging: $10I = 7A - A^2 \Rightarrow I = \frac{1}{10}(7A - A^2) = A \cdot \frac{1}{10}(7I - A)$. Therefore: $A^{-1} = \frac{1}{10}(7I - A) = \frac{1}{10}\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 2/5 & -1/10 \\ -1/5 & 3/10 \end{pmatrix}$. Verify: $\det(A) = 12 - 2 = 10$, so $A^{-1} = \frac{1}{10}\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}$ ✓.

Matrix Multiplication — Properties and Examples

Key properties:

- Matrix multiplication is **not** commutative in general: $AB \neq BA$.
- Associative: $(AB)C = A(BC)$.
- Distributive: $A(B + C) = AB + AC$.
- $(AB)^T = B^T A^T$, $(AB)^{-1} = B^{-1}A^{-1}$.
- For square A: $\det(AB) = \det(A)\det(B)$.

Example: Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. Find AB and BA to show they differ.
$AB = \begin{pmatrix} 4 & 7 \\ 8 & 15 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 4 \\ 11 & 16 \end{pmatrix}$. Indeed $AB \neq BA$.

Matrix Powers Using Diagonalisation

Example: $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. Find A^n.

Eigenvalues: $\lambda = 3, \lambda = 2$.

For $\lambda = 3$: $\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}\mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

For $\lambda = 2$: $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, $P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
$A^n = PD^nP^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3^n & 3^n - 2^n \\ 0 & 2^n \end{pmatrix}$.

Inverse of a 3×3 Matrix

For a 3×3 matrix A: $A^{-1} = \dfrac{1}{\det A} \operatorname{adj}(A)$ where $\operatorname{adj}(A)$ is the transpose of the matrix of cofactors.
Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$ where M_{ij} is the minor (determinant of the 2×2 matrix with row i and column j removed).

Example: $A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$.
$\det A = 1(1 \cdot 1 - 3 \cdot 0) - 2((-1) \cdot 1 - 3 \cdot 2) + 0 = 1 + 14 = 15$.
Cofactors $C_{11} = 1,\ C_{12} = 7,\ C_{13} = -2,\ C_{21} = -2,\ C_{22} = 1,\ C_{23} = 4,\ C_{31} = 6,\ C_{32} = -3,\ C_{33} = 3$.
$\operatorname{adj}(A) = \begin{pmatrix} 1 & -2 & 6 \\ 7 & 1 & -3 \\ -2 & 4 & 3 \end{pmatrix}$,\quad $A^{-1} = \dfrac{1}{15} \begin{pmatrix} 1 & -2 & 6 \\ 7 & 1 & -3 \\ -2 & 4 & 3 \end{pmatrix}$.

Linear Transformations — Composition

Applying transformation T_1 (matrix A) then T_2 (matrix B) is represented by BA (note the reversed order).
The image of point $\mathbf{x}$ under transformation T is $A\mathbf{x}$.

Example: Reflect in the x-axis ($A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$) then rotate 90° anticlockwise ($B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$).
Combined transformation:

$BA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ — reflection in $y = x$.

Example — Stretch and rotate: Scale by 2 then rotate 45°:
$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$.
$BA = \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}$. This represents an enlargement by 2 combined with a rotation.

Matrices — AQA Quick Reference

Transformation	Matrix
Rotation by θ (anticlockwise)	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
Reflection in $y = x\tan\alpha$	$\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$
Enlargement scale factor k	$k\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Stretch in x by a, y by b	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$
Shear (parallel to x-axis)	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Characteristic Polynomial and Minimal Polynomial

The **characteristic polynomial** of A is $\chi_A(\lambda) = \det(\lambda I - A)$.
The **minimal polynomial** $m_A(\lambda)$ is the monic polynomial of lowest degree such that $m_A(A) = O$.
By Cayley-Hamilton: $\chi_A(A) = O$. The minimal polynomial divides the characteristic polynomial.

Example: For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Characteristic polynomial: $(\lambda - 1)^2$. Both eigenvalues are 1.

Is $m_A = (\lambda - 1)$? Test: $A - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq O$. So minimal polynomial is $(\lambda - 1)^2$.

Example 2: For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$.

Characteristic polynomial: $(\lambda - 2)^2$. But $A - 2I = O$, so minimal polynomial is $(\lambda - 2)$.

AQA Matrices Exam Technique:

- For eigenvalue/eigenvector questions: always verify by computing $A\mathbf{v} = \lambda\mathbf{v}$ for your final answer.
- For Cayley-Hamilton: substitute A for λ in the characteristic equation — be careful with $\lambda^0 = I$ (not 1).
- To find A^{-1} via Cayley-Hamilton: rearrange so that $I = f(A)$ and read off A^{-1}.
- Diagonalisation: $A^n = PD^nP^{-1}$ only works when A has n linearly independent eigenvectors (i.e. is diagonalisable).
- For a 2×2 matrix: if $\mathrm{tr}(A) = \lambda_1 + \lambda_2$ and $\det(A) = \lambda_1\lambda_2$, these give a quick check on your eigenvalues.
- Eigenvectors are not unique — any non-zero scalar multiple is also an eigenvector. In AQA, state the direction clearly or write the general form $k\mathbf{v}$.
- Row operations: when solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$, use row reduction. The rows should be linearly dependent (since $\det = 0$), so one equation is redundant — use the remaining equation to express components in terms of a free parameter.

Determinant Properties — Reminder

- $\det(AB) = \det(A)\det(B)$
- $\det(A^{-1}) = 1/\det(A)$

- $\det(A^T) = \det(A)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix
- Swapping two rows (or columns) multiplies the determinant by -1.
- Adding a multiple of one row to another does not change the determinant.